Packages

Module-2



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Packages

- Java package is a mechanism of grouping similar type of classes, interfaces, and sub-classes collectively based on functionality.
- When software is written in the Java programming language, it can be composed of hundreds or even thousands of individual classes.
- It makes sense to keep things organized by placing related classes and interfaces into packages.

```
import java.util.Scanner
```

Packages

key points

- Every class belongs to a package. If you do not mention any package, the classes in the file move into a special unnamed package which is the same for all the files which do not belong to a specified package.
- Multiple classes and interfaces in a file can be part of the same package.
- If the package name is specified, then the directory name must match the package name.
- We can access the classes declared as public in another package using: import package-name.class-name

Packages

key points

- Organize your classes into smaller units and make it easy to locate and use the appropriate class file
- Avoid naming conflicts

    - Package names can be used to identify your classes

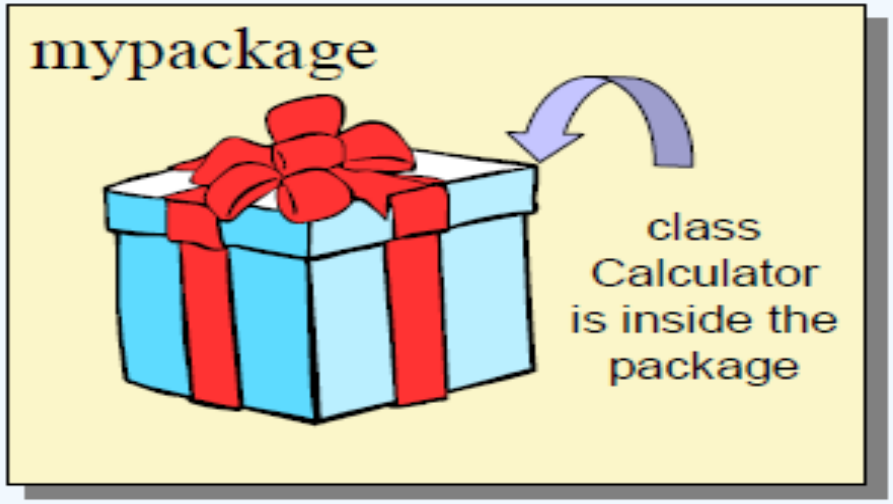
- Packages allow you to protect your classes, data and methods in a larger way than on a class-to-class basis

Packages

Creating a Package

In Java Packages are created in the following manner :

```
package package_name ;
```




```
package mypackage ;
public class Calculator
{
public int add(int x, int y)
{
    return( x + y ) ;
}
}
```

Packages

Importing Packages

- **Importing a specific package member**

  - import mypackage.Calculator

- **Importing an Entire Package**

  - import mypackage.*;

- **Importing + Creating a Package**

  While **Importing an Entire Package**, care should be taken that the statement for creating a package must be written before any other import statements

  Ex: package mypackage;
  import java.io.*;

Packages

Sub-Packages

- We can also put a package inside an another package
- The packages that comes lower in the naming hierarchy are called "subpackage" of the corresponding package higher in the hierarchy.
- Place the sub-package folder within main package directory.

```
package mypackage.subpackage;
```

Packages

Sub-Packages: Creation

```
package mypackage.subpackage;
public class HelloWorld {
  public void show(){
  System.out.println("This is the function of the class HelloWorld!!");
  }
}
```

Packages

Sub-Packages: Importing in main program

```
import mypackage.subpackage.*;
class CallPackage{
  public static void main(String[] args){
  HelloWorld h2=new HelloWorld();
  h2.show();
  }
}
```

Packages

Benefits

Resolve naming conflict



Reuse of code



Organization of project





Modularity



Access protection



Information hiding